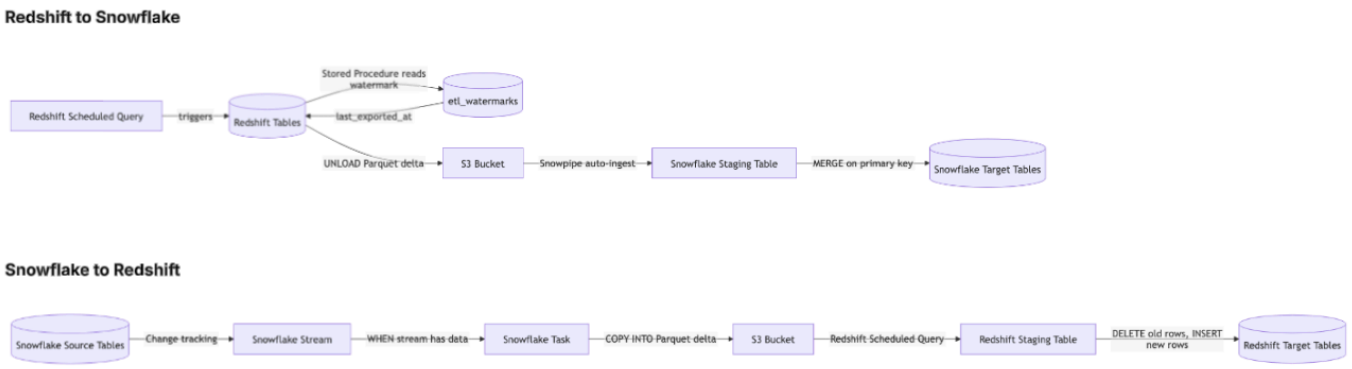
Redshift ↔ Snowflake Bidirectional Integration: Implementation Guide

Overview

This guide covers a bidirectional data integration between AWS Redshift and Snowflake using S3 as the interchange layer. No Airflow, Step Functions, or third-party ETL tools are required.



Assumptions

	Redshift → Snowflake	Snowflake → Redshift
Number of tables	15	1
Initial bulk load	~10M rows per table	~2M rows
Ongoing	Incremental (changed rows only)	Incremental (CDC via stream)
Sync frequency	Weekly	Weekly

The watermark-based incremental procedure on the Redshift side means only rows changed since the last run are exported — not the full 10M rows — after the initial bulk load completes. Similarly, the Snowflake stream only captures changed rows after the initial task execution.

Two independent data flows:

Direction	Source Table	Target Table	Trigger
Redshift → Snowflake	`shipments` (Redshift)	`shipments` (Snowflake)	Redshift Stored Procedure + Scheduler
Snowflake → Redshift	`inventory_forecast` (Snowflake)	`inventory_forecast` (Redshift)	Snowflake Stream + Task (CDC)

Ownership boundary: Each system owns its source table. Neither system writes to the other's source — conflict-free by design.

> **Note on table names:** This guide uses `shipments` and `inventory_forecast` as example table names. Replace these with your own table names throughout.

Prerequisites

Before starting, fill in these values for your environment. They are referenced throughout this guide as placeholders in `< >`.

AWS

Placeholder	Description	Where to Find
`<your-aws-account-id>`	12-digit AWS account number	AWS console top-right menu
`<your-aws-region>`	AWS region for all resources	e.g. `us-east-1`
`<your-redshift-cluster>`	Redshift cluster identifier	Redshift console
`<your-redshift-database>`	Database name inside the cluster	e.g. `dev` or `prod`
`<your-db-user>`	Redshift user for Data API calls	e.g. `admin`
`<your-s3-bucket>`	S3 bucket used as the interchange layer	S3 console
`<your-redshift-s3-role>`	IAM role name attached to Redshift cluster	IAM console
`<your-eventbridge-role>`	IAM role name for EventBridge Scheduler	IAM console (created in this guide)
`<your-scheduler-name>`	EventBridge Scheduler name	Your choice, e.g. `redshift-incremental-export`

Snowflake

Placeholder	Description	Where to Find
`<your-snowflake-account>`	Snowflake account identifier	Admin → Accounts
`<your-snowflake-database>`	Target database	e.g. `ANALYTICS_DB`
`<your-snowflake-schema>`	Target schema	e.g. `REDSHIFT_INTEGRATION`
`<your-snowflake-role>`	Role used for setup	e.g. `SYSADMIN`

Placeholder	Description	Where to Find
`<your-snowflake-warehouse>`	Warehouse for task execution	e.g. `COMPUTE_WH`
`<your-storage-integration>`	Storage integration name	Your choice, e.g. `s3_integration`
`<your-stage>`	External stage name	Your choice, e.g. `s3_stage`
`<snowflake-iam-user-arn>`	Snowflake's IAM user ARN	`DESC INTEGRATION <your-storage-integration>`
`<snowflake-external-id>`	Snowflake external ID for trust	`DESC INTEGRATION <your-storage-integration>`
`<sqs-arn-from-show-pipes>`	SQS ARN for Snowpipe auto-ingest	`SHOW PIPES` after creating the pipe

S3 Path Convention

This guide uses the following S3 path structure. Adjust the prefix to match your naming conventions.

Path	Purpose
`s3://<your-s3-bucket>/redshift-to-snowflake/shipments/<timestamp>/`	Incremental UNLOAD from Redshift
`s3://<your-s3-bucket>/redshift-to-snowflake/shipments_bulk/`	One-time bulk export from Redshift
`s3://<your-s3-bucket>/snowflake-to-redshift/inventory_forecast/`	Stream export from Snowflake

Flow 1: Redshift → Snowflake (`shipments`)

Architecture

```
Redshift: shipments
    │
    │  UNLOAD delta (updated_at > watermark)
    ▼
S3: redshift-to-snowflake/shipments/<YYYY-MM-DD-HH24MISS>/
    │
    │  Snowpipe AUTO_INGEST (SQS event notification)
    ▼
Snowflake: shipments_staging
    │
    │  WHEN SYSTEM$STREAM_HAS_DATA() (hourly cron)
    ▼
merge_shipments_task
    │
```

```
    │  MERGE with ROW_NUMBER() dedup
    ▼
Snowflake: shipments (target)

Scheduler ──► CALL export_shipments_incremental()
```

Redshift Objects

Watermark tracking table

```sql
CREATE TABLE IF NOT EXISTS etl_watermarks (
    table_name       VARCHAR(255),
    last_exported_at TIMESTAMP,
    updated_at       TIMESTAMP DEFAULT SYSDATE
);

-- Initialize once before first run.
-- Set to a date before your earliest data to capture full history on first
run.
INSERT INTO etl_watermarks (table_name, last_exported_at)
VALUES ('shipments', '2020-01-01 00:00:00');
```

Source table

```sql
CREATE TABLE IF NOT EXISTS shipments (
    shipment_id        INT,
    order_id           VARCHAR(20),
    supplier_id        VARCHAR(20),
    origin_warehouse   VARCHAR(50),
    destination        VARCHAR(100),
    carrier            VARCHAR(50),
    status             VARCHAR(20),
    ship_date          DATE,
    estimated_arrival  DATE,
    actual_arrival     DATE,
    quantity           INT,
    weight_kg          DECIMAL(8,2),
    updated_at         TIMESTAMP DEFAULT SYSDATE
);
```

Stored procedure for incremental UNLOAD

```sql
CREATE OR REPLACE PROCEDURE export_shipments_incremental()
AS $$
DECLARE
    v_last_run TIMESTAMP;
```

```
    v_this_run TIMESTAMP;
    v_ts_path  VARCHAR(50);
    v_sql      VARCHAR(65535);
BEGIN
    -- Capture current time at START of run (not end).
    -- This ensures rows written mid-export are not missed next run.
    v_this_run := SYSDATE;

    -- Build timestamp string for S3 path (makes every run unique).
    v_ts_path  := TO_CHAR(v_this_run, 'YYYY-MM-DD-HH24MISS');

    -- Read last successful export timestamp.
    SELECT last_exported_at INTO v_last_run
    FROM etl_watermarks
    WHERE table_name = 'shipments';

    -- Build and execute incremental UNLOAD.
    v_sql := 'UNLOAD (''SELECT * FROM shipments WHERE updated_at > ''''' ||
v_last_run || '''''')
              TO ''s3://<your-s3-bucket>/redshift-to-snowflake/shipments/'
|| v_ts_path || '/''
              IAM_ROLE ''arn:aws:iam::<your-aws-account-id>:role/<your-
redshift-s3-role>''
              FORMAT AS PARQUET ALLOWOVERWRITE';

    EXECUTE v_sql;

    -- Update watermark only after successful UNLOAD.
    -- If the procedure fails, watermark stays at the last checkpoint.
    UPDATE etl_watermarks
    SET last_exported_at = v_this_run,
        updated_at       = SYSDATE
    WHERE table_name = 'shipments';
END;
$$ LANGUAGE plpgsql;
```

> **Why timestamped S3 paths?** Snowpipe deduplicates by filename. If the procedure always writes to `shipments/0000_part_00.parquet`, Snowpipe skips it after the first ingest. Writing to `shipments/2026-04-24-000119/0000_part_00.parquet` makes every run produce a unique path that Snowpipe always ingests.

Bulk UNLOAD (Phase 1 — run once)

```
UNLOAD ('SELECT * FROM shipments')
TO 's3://<your-s3-bucket>/redshift-to-snowflake/shipments_bulk/'
IAM_ROLE 'arn:aws:iam::<your-aws-account-id>:role/<your-redshift-s3-role>'
FORMAT AS PARQUET
ALLOWOVERWRITE;

-- Reset watermark after bulk load so incremental runs only pick up new
rows.
```

```
UPDATE etl_watermarks
SET last_exported_at = SYSDATE,
    updated_at       = SYSDATE
WHERE table_name = 'shipments';
```

Snowflake Objects

Tables

```
-- Target table (receives from Redshift)
CREATE OR REPLACE TABLE <your-snowflake-database>.<your-snowflake-
schema>.shipments (
    shipment_id       NUMBER,
    order_id          VARCHAR(20),
    supplier_id       VARCHAR(20),
    origin_warehouse  VARCHAR(50),
    destination       VARCHAR(100),
    carrier           VARCHAR(50),
    status            VARCHAR(20),
    ship_date         DATE,
    estimated_arrival DATE,
    actual_arrival    DATE,
    quantity          NUMBER,
    weight_kg         NUMBER(8,2),
    updated_at        TIMESTAMP_NTZ
);

-- Staging table for Snowpipe ingest (identical schema to target)
CREATE OR REPLACE TABLE <your-snowflake-database>.<your-snowflake-
schema>.shipments_staging (
    shipment_id       NUMBER,
    order_id          VARCHAR(20),
    supplier_id       VARCHAR(20),
    origin_warehouse  VARCHAR(50),
    destination       VARCHAR(100),
    carrier           VARCHAR(50),
    status            VARCHAR(20),
    ship_date         DATE,
    estimated_arrival DATE,
    actual_arrival    DATE,
    quantity          NUMBER,
    weight_kg         NUMBER(8,2),
    updated_at        TIMESTAMP_NTZ
);
```

Snowpipe

```
CREATE OR REPLACE PIPE <your-snowflake-database>.<your-snowflake-
schema>.shipments_pipe
  AUTO_INGEST = TRUE
AS
COPY INTO <your-snowflake-database>.<your-snowflake-
schema>.shipments_staging
FROM @<your-snowflake-database>.<your-snowflake-schema>.<your-
stage>/redshift-to-snowflake/shipments/
FILE_FORMAT = (TYPE = PARQUET)
MATCH_BY_COLUMN_NAME = CASE_INSENSITIVE;

-- Retrieve SQS ARN — needed for S3 event notification setup (next step)
SHOW PIPES LIKE 'shipments_pipe' IN SCHEMA <your-snowflake-database>.<your-
snowflake-schema>;
-- Copy the value in the notification_channel column: <sqs-arn-from-show-
pipes>
```

After creating the pipe, configure S3 to notify Snowpipe when new files arrive:

1. Go to the **S3 console** → open `<your-s3-bucket>` → **Properties** tab
2. Scroll to **Event notifications** → click **Create event notification**
3. Fill in:
   - **Event name:** anything descriptive, e.g. `snowpipe-shipments-trigger`
   - **Prefix:** `redshift-to-snowflake/shipments/` (must match the pipe's S3 path exactly)
   - **Event types:** check `s3:ObjectCreated:*`
   - **Destination:** select **SQS queue** → **Enter SQS queue ARN** → paste the ARN from `SHOW PIPES`
4. Click **Save changes**

To verify: UNLOAD a file to the prefix and check Snowpipe ingest history in Snowflake:

```
SELECT * FROM TABLE(INFORMATION_SCHEMA.COPY_HISTORY(
    TABLE_NAME => 'shipments_staging',
    START_TIME => DATEADD('hour', -1, CURRENT_TIMESTAMP)
));
```

Bulk COPY (Phase 1 — run once)

```
-- Copy directly into the TARGET table (not staging).
-- The bulk Parquet has no METADATA$ columns — staging schema matches, but
-- keeping bulk loads in the target avoids confusion with the incremental
flow.
COPY INTO <your-snowflake-database>.<your-snowflake-schema>.shipments
FROM @<your-snowflake-database>.<your-snowflake-schema>.<your-
stage>/redshift-to-snowflake/shipments_bulk/
```

```
  FILE_FORMAT = (TYPE = PARQUET)
  MATCH_BY_COLUMN_NAME = CASE_INSENSITIVE;
```

Stream (on staging table)

```
CREATE OR REPLACE STREAM <your-snowflake-database>.<your-snowflake-
schema>.shipments_staging_stream
  ON TABLE <your-snowflake-database>.<your-snowflake-
schema>.shipments_staging
  APPEND_ONLY = TRUE;
```

`APPEND_ONLY = TRUE` because Snowpipe only ever inserts into staging — no updates or deletes. The stream tracks exactly which rows Snowpipe has added since the last task run.

Automated MERGE task

```
CREATE OR REPLACE TASK <your-snowflake-database>.<your-snowflake-
schema>.merge_shipments_task
  WAREHOUSE = <your-snowflake-warehouse>
  SCHEDULE  = 'USING CRON 0 * * * * UTC'   -- hourly; adjust for production
WHEN
  SYSTEM$STREAM_HAS_DATA('<your-snowflake-database>.<your-snowflake-
schema>.shipments_staging_stream')
AS
MERGE INTO <your-snowflake-database>.<your-snowflake-schema>.shipments AS
tgt
USING (
    SELECT * FROM (
        SELECT *,
               ROW_NUMBER() OVER (PARTITION BY shipment_id ORDER BY
updated_at DESC) AS rn
        FROM <your-snowflake-database>.<your-snowflake-
schema>.shipments_staging_stream
    ) WHERE rn = 1
) AS src
ON tgt.shipment_id = src.shipment_id
WHEN MATCHED THEN UPDATE SET
    order_id          = src.order_id,
    status            = src.status,
    actual_arrival    = src.actual_arrival,
    updated_at        = src.updated_at
WHEN NOT MATCHED THEN INSERT
    (shipment_id, order_id, supplier_id, origin_warehouse, destination,
carrier, status,
     ship_date, estimated_arrival, actual_arrival, quantity, weight_kg,
updated_at)
VALUES
    (src.shipment_id, src.order_id, src.supplier_id, src.origin_warehouse,
```

```
src.destination,
     src.carrier, src.status, src.ship_date, src.estimated_arrival,
src.actual_arrival,
     src.quantity, src.weight_kg, src.updated_at);

-- Tasks are suspended by default after creation — resume before use
ALTER TASK <your-snowflake-database>.<your-snowflake-
schema>.merge_shipments_task RESUME;

-- Manually trigger for testing (bypasses the schedule)
EXECUTE TASK <your-snowflake-database>.<your-snowflake-
schema>.merge_shipments_task;
```

> **How this works end-to-end:** Snowpipe writes new rows to `shipments_staging` → the stream captures those inserts → the task fires (at most hourly) when `SYSTEM$STREAM_HAS_DATA()` is true → MERGE upserts the delta into the target → stream offset advances so those rows are never reprocessed.
>
> `ROW_NUMBER()` deduplication handles the case where Snowpipe ingests the same row more than once (e.g., from overlapping incremental windows). Without it, Snowflake throws "Duplicate row detected during DML action".
>
> **Adapting this MERGE for your table:** Two things need to change:
>
> 1. Replace `shipment_id` in `PARTITION BY` and the `ON` clause with your table's primary key column
> 2. In `WHEN MATCHED UPDATE SET`, list only the columns that can change on your table — omit immutable columns like IDs and `created_at`
>
> The structure of the MERGE itself (the `ROW_NUMBER()` wrapper, `WHEN MATCHED / WHEN NOT MATCHED`) is the same for every table.

Scheduling the Stored Procedure

Option A — Redshift Query Editor v2 (recommended, no IAM setup required):

1. Open Query Editor v2 in the Redshift console
2. Write: `CALL export_shipments_incremental();`
3. Save the query
4. Click **Schedule** → set frequency → save

Option B — EventBridge Scheduler (CLI):

```
# Step 1: Create IAM role for EventBridge to call the Redshift Data API
aws iam create-role \
  --role-name <your-eventbridge-role> \
  --assume-role-policy-document '{
    "Version": "2012-10-17",
    "Statement": [{
      "Effect": "Allow",
```

```
      "Principal": {"Service": "scheduler.amazonaws.com"},
      "Action": "sts:AssumeRole",
      "Condition": {
        "StringEquals": {"aws:SourceAccount": "<your-aws-account-id>"}
      }
    }]
  }'

# IMPORTANT: The aws:SourceAccount condition is required.
# Without it, the scheduler is created successfully but fires 0
invocations.

# Step 2: Attach permission to call Redshift Data API
aws iam put-role-policy \
  --role-name <your-eventbridge-role> \
  --policy-name redshift-data-invoke \
  --policy-document '{
    "Version": "2012-10-17",
    "Statement": [{
      "Effect": "Allow",
      "Action": [
        "redshift-data:ExecuteStatement",
        "redshift:GetClusterCredentials"
      ],
      "Resource": "*"
    }]
  }'

# Step 3: Create the schedule (adjust rate or cron as needed)
aws scheduler create-schedule \
  --name <your-scheduler-name> \
  --schedule-expression "cron(0 6 ? * MON *)" \
  --flexible-time-window '{"Mode": "OFF"}' \
  --target '{
    "Arn": "arn:aws:scheduler:::aws-sdk:redshiftdata:executeStatement",
    "RoleArn": "arn:aws:iam::<your-aws-account-id>:role/<your-eventbridge-
role>",
    "Input": "{\"ClusterIdentifier\":\"<your-redshift-
cluster>\",\"Database\":\"<your-redshift-database>\",\"DbUser\":\"<your-db-
user>\",\"Sql\":\"CALL export_shipments_incremental();\"}"
  }' \
  --region <your-aws-region>
```

Flow 2: Snowflake → Redshift (`inventory_forecast`)

> **Alternative approach:** Instead of Snowflake Streams (CDC), some teams use a watermark pattern symmetric with Flow 1 — Redshift stores a `last_exported_at` timestamp per table, Snowflake queries `WHERE updated_at > watermark` and UNLOADs to S3, Redshift COPYs in. This is simpler and consistent with the Redshift → Snowflake pattern, but does not capture deletes. If your tables are append/update only with no row deletions, this may be the better choice. The stream-based approach below handles all change types including deletes.

Architecture

```
Snowflake: inventory_forecast
    |
    |  Change data capture
    ▼
Snowflake: inventory_forecast_stream
    |
    |  WHEN SYSTEM$STREAM_HAS_DATA() (weekly cron)
    ▼
export_inventory_forecast_task
    |
    |  COPY INTO Parquet (OVERWRITE=TRUE, explicit type casts, aliased
METADATA$ columns)
    ▼
S3: snowflake-to-redshift/inventory_forecast/
    |
    |  Redshift scheduled COPY (runs shortly after task)
    ▼
Redshift: inventory_forecast_staging (raw CDC rows)
    |
    |  DELETE old/updated rows, INSERT new rows → target
    ▼
Redshift: inventory_forecast (target, business columns only)
```

Snowflake Objects

Source table

```sql
CREATE OR REPLACE TABLE <your-snowflake-database>.<your-snowflake-
schema>.inventory_forecast (
    forecast_id      NUMBER,
    sku_id           VARCHAR(20),
    product_name     VARCHAR(100),
    warehouse_id     VARCHAR(20),
    forecast_date    DATE,
    forecasted_qty   NUMBER,
    actual_qty       NUMBER,
    confidence_score NUMBER(5,4),
    model_version    VARCHAR(20),
    updated_at       TIMESTAMP_NTZ
);
```

Stream

```sql
CREATE OR REPLACE STREAM <your-snowflake-database>.<your-snowflake-
schema>.inventory_forecast_stream
```

```
  ON TABLE <your-snowflake-database>.<your-snowflake-schema>.inventory_forecast
  APPEND_ONLY = FALSE;
```

`APPEND_ONLY = FALSE` captures inserts, updates, AND deletes. Updates appear as two rows: a `DELETE` (old values) followed by an `INSERT` (new values).

The stream adds 3 metadata columns that must be carried through to the Redshift target:

Column	Values	Meaning
`METADATA$ACTION`	`INSERT` / `DELETE`	Type of change
`METADATA$ISUPDATE`	`TRUE` / `FALSE`	Part of an update operation
`METADATA$ROW_ID`	string	Unique internal row identifier

Task

```
CREATE OR REPLACE TASK <your-snowflake-database>.<your-snowflake-schema>.export_inventory_forecast_task
  WAREHOUSE = <your-snowflake-warehouse>
  SCHEDULE  = 'USING CRON 0 * * * * UTC'   -- hourly; adjust for production
WHEN
  SYSTEM$STREAM_HAS_DATA('<your-snowflake-database>.<your-snowflake-schema>.inventory_forecast_stream')
AS
COPY INTO @<your-snowflake-database>.<your-snowflake-schema>.<your-stage>/snowflake-to-redshift/inventory_forecast/
FROM (
    SELECT
        forecast_id::BIGINT           AS forecast_id,
        sku_id::VARCHAR(20)           AS sku_id,
        product_name::VARCHAR(100)    AS product_name,
        warehouse_id::VARCHAR(20)     AS warehouse_id,
        forecast_date::TIMESTAMP_NTZ  AS forecast_date,
        forecasted_qty::BIGINT        AS forecasted_qty,
        actual_qty::BIGINT            AS actual_qty,
        confidence_score::NUMBER(5,4) AS confidence_score,
        model_version::VARCHAR(20)    AS model_version,
        updated_at::TIMESTAMP_NTZ     AS updated_at,
        METADATA$ACTION               AS metadata_action,
        METADATA$ISUPDATE             AS metadata_isupdate,
        METADATA$ROW_ID               AS metadata_row_id
    FROM <your-snowflake-database>.<your-snowflake-schema>.inventory_forecast_stream
)
FILE_FORMAT = (TYPE = PARQUET)
OVERWRITE   = TRUE
HEADER      = TRUE;
```

```
-- Tasks are suspended by default after creation — resume before use
ALTER TASK <your-snowflake-database>.<your-snowflake-
schema>.export_inventory_forecast_task RESUME;

-- Manually trigger for testing (bypasses the schedule)
EXECUTE TASK <your-snowflake-database>.<your-snowflake-
schema>.export_inventory_forecast_task;
```

> **Why explicit type casts?** Snowflake `NUMBER` columns write as `decimal128(38,0)` in Parquet. Redshift cannot load `decimal128(38,0)` into `BIGINT` or `INT` columns — it throws a schema incompatibility error. Casting to `::BIGINT` in the task SELECT writes `int64` in Parquet, which Redshift accepts for `BIGINT` columns. See the type mapping table below.
>
> **Why alias METADATA$ columns?** The `$` character in `METADATA$ACTION` is not valid in Redshift column names. Aliasing as `metadata_action` (no `$`) resolves this.
>
> **Why `OVERWRITE = TRUE`?** The task writes to a fixed S3 path on every run. Without `OVERWRITE = TRUE`, Snowflake rejects the COPY if files already exist at the destination.

Bulk load (Phase 1 — run once)

After inserting your initial data, execute the task manually. The stream captures the initial inserts and the task writes them to S3.

```
EXECUTE TASK <your-snowflake-database>.<your-snowflake-
schema>.export_inventory_forecast_task;
```

Redshift Objects

Target table

```
-- IMPORTANT: Use BIGINT for integer columns (see type mapping section).
-- IMPORTANT: Include all three metadata columns — the Parquet always
contains them.
CREATE TABLE IF NOT EXISTS inventory_forecast (
    forecast_id        BIGINT,
    sku_id             VARCHAR(20),
    product_name       VARCHAR(100),
    warehouse_id       VARCHAR(20),
    forecast_date      TIMESTAMP,
    forecasted_qty     BIGINT,
    actual_qty         BIGINT,
    confidence_score   DECIMAL(5,4),
    model_version      VARCHAR(20),
    updated_at         TIMESTAMP,
    metadata_action    VARCHAR(10),
    metadata_isupdate  BOOLEAN,
```

```
    metadata_row_id    VARCHAR(100)
);
```

Staging table

> **Note on implementation:** The staging + DELETE+INSERT pattern below is the recommended production approach for keeping Redshift as an exact mirror of Snowflake. For initial testing and validation, a simple `COPY` directly into the target table is sufficient — but it will not correctly handle updates or deletes, and the target table will include the three metadata columns. Switch to the full pattern below before going to production.

For production, create a staging table with the same schema as the target (including the three metadata columns). The staging table acts as a landing zone for the raw CDC Parquet — the target table stays clean with business columns only after applying changes.

```sql
CREATE TABLE IF NOT EXISTS inventory_forecast_staging (
    forecast_id        BIGINT,
    sku_id             VARCHAR(20),
    product_name       VARCHAR(100),
    warehouse_id       VARCHAR(20),
    forecast_date      TIMESTAMP,
    forecasted_qty     BIGINT,
    actual_qty         BIGINT,
    confidence_score   DECIMAL(5,4),
    model_version      VARCHAR(20),
    updated_at         TIMESTAMP,
    metadata_action    VARCHAR(10),
    metadata_isupdate  BOOLEAN,
    metadata_row_id    VARCHAR(100)
);
```

COPY and apply changes

```sql
-- Step 1: Load CDC delta into staging
COPY inventory_forecast_staging
FROM 's3://<your-s3-bucket>/snowflake-to-redshift/inventory_forecast/'
IAM_ROLE 'arn:aws:iam::<your-aws-account-id>:role/<your-redshift-s3-role>'
FORMAT AS PARQUET;

-- Step 2: Remove deleted and updated rows from target
-- (Snowflake streams represent updates as a DELETE row + INSERT row pair,
-- so this handles both deletes and the old version of updates)
DELETE FROM inventory_forecast
USING inventory_forecast_staging s
WHERE inventory_forecast.forecast_id = s.forecast_id
  AND s.metadata_action = 'DELETE';

-- Step 3: Insert new and updated rows into target
```

```
-- NOTE: We carry the three metadata columns through to the target here for
-- transparency and debugging. In a production deployment you would
typically
-- strip them and keep the target to business columns only — remove
-- metadata_action, metadata_isupdate, metadata_row_id from both the INSERT
-- column list and the SELECT below, and drop those columns from the target
table.
INSERT INTO inventory_forecast (
    forecast_id, sku_id, product_name, warehouse_id, forecast_date,
    forecasted_qty, actual_qty, confidence_score, model_version,
updated_at,
    metadata_action, metadata_isupdate, metadata_row_id
)
SELECT
    forecast_id, sku_id, product_name, warehouse_id, forecast_date,
    forecasted_qty, actual_qty, confidence_score, model_version,
updated_at,
    metadata_action, metadata_isupdate, metadata_row_id
FROM inventory_forecast_staging
WHERE metadata_action = 'INSERT';
```

> **Why DELETE + INSERT instead of UPDATE?** Snowflake streams do not emit UPDATE rows. An update in Snowflake produces two rows in the stream: a `DELETE` with the old values and an `INSERT` with the new values. Processing all DELETEs then all INSERTs applies updates correctly without needing a SQL UPDATE statement.
>
> **Scheduling:** Wrap the COPY + apply block in the `copy_inventory_forecast()` stored procedure below and schedule it via EventBridge Scheduler. The Snowflake task fires at `:00` UTC each hour; schedule the Redshift procedure at `:30` to guarantee the Parquet file has been written before the COPY starts.

Stored procedure — COPY + apply CDC

```
CREATE OR REPLACE PROCEDURE copy_inventory_forecast()
AS $$
BEGIN
    TRUNCATE TABLE <your-redshift-schema>.inventory_forecast_staging;

    EXECUTE 'COPY <your-redshift-schema>.inventory_forecast_staging
    FROM ''s3://<your-s3-bucket>/snowflake-to-
redshift/inventory_forecast/''
    IAM_ROLE ''arn:aws:iam::<your-aws-account-id>:role/<your-redshift-s3-
role>''
    FORMAT AS PARQUET';

    DELETE FROM <your-redshift-schema>.inventory_forecast
    USING <your-redshift-schema>.inventory_forecast_staging AS s
    WHERE inventory_forecast.forecast_id = s.forecast_id
      AND s.metadata_action IN ('DELETE', 'UPDATE_PRE');
```

```
    INSERT INTO <your-redshift-schema>.inventory_forecast (
        forecast_id, sku_id, product_name, warehouse_id,
        forecast_date, forecasted_qty, actual_qty, confidence_score,
        model_version, updated_at,
        metadata_action, metadata_isupdate, metadata_row_id
    )
    SELECT
        forecast_id, sku_id, product_name, warehouse_id,
        forecast_date, forecasted_qty, actual_qty, confidence_score,
        model_version, updated_at,
        metadata_action, metadata_isupdate, metadata_row_id
    FROM <your-redshift-schema>.inventory_forecast_staging
    WHERE metadata_action IN ('INSERT', 'UPDATE_POST');
END;
$$ LANGUAGE plpgsql;
```

EventBridge Scheduler — automate the COPY

Create an EventBridge Scheduler schedule that calls the procedure via the Redshift Data API. The Snowflake task runs at `:00` UTC; schedule this at `:30` to give the task time to finish.

```
aws scheduler create-schedule \
  --name "<your-scheduler-name>" \
  --schedule-expression "cron(30 * * * ? *)" \
  --schedule-expression-timezone "UTC" \
  --flexible-time-window '{"Mode": "OFF"}' \
  --target '{
    "Arn": "arn:aws:scheduler:::aws-sdk:redshiftdata:executeStatement",
    "RoleArn": "arn:aws:iam::<your-aws-account-id>:role/<your-eventbridge-
role>",
    "Input": "{\"ClusterIdentifier\":\"<your-redshift-
cluster>\",\"Database\":\"<your-redshift-database>\",\"DbUser\":\"<your-db-
user>\",\"Sql\":\"CALL
copy_inventory_forecast();\",\"StatementName\":\"eventbridge-copy-
inventory-forecast\"}"
  }' \
  --region <your-aws-region>
```

> **Why `arn:aws:scheduler:::aws-sdk:redshiftdata:executeStatement`?** EventBridge Scheduler uses AWS SDK universal targets (not `RedshiftDataParameters`). The ARN prefix `arn:aws:scheduler:::aws-sdk:` maps directly to the AWS SDK service and operation you want to invoke.

IAM Configuration

Redshift → S3 (`<your-redshift-s3-role>`)

Trust policy — allows Redshift to assume this role:

```
{
  "Version": "2012-10-17",
  "Statement": [{
    "Effect": "Allow",
    "Principal": {"Service": "redshift.amazonaws.com"},
    "Action": "sts:AssumeRole"
  }]
}
```

Permission policy — S3 read/write on the interchange bucket:

```
{
  "Version": "2012-10-17",
  "Statement": [{
    "Effect": "Allow",
    "Action": ["s3:GetObject", "s3:PutObject", "s3:ListBucket",
"s3:DeleteObject"],
    "Resource": [
      "arn:aws:s3:::<your-s3-bucket>",
      "arn:aws:s3:::<your-s3-bucket>/*"
    ]
  }]
}
```

After creating the role, attach it to your Redshift cluster under **Properties → IAM roles**.

Snowflake → S3 (Storage Integration role)

Step 1 — Create the Storage Integration in Snowflake:

```
CREATE OR REPLACE STORAGE INTEGRATION <your-storage-integration>
  TYPE = EXTERNAL_STAGE
  STORAGE_PROVIDER = 'S3'
  ENABLED = TRUE
  STORAGE_AWS_ROLE_ARN = 'arn:aws:iam::<your-aws-account-id>:role/<your-
snowflake-s3-role>'
  STORAGE_ALLOWED_LOCATIONS = ('s3://<your-s3-bucket>/');

-- Retrieve the values needed for the IAM trust policy
DESC INTEGRATION <your-storage-integration>;
-- Copy: STORAGE_AWS_IAM_USER_ARN  → <snowflake-iam-user-arn>
-- Copy: STORAGE_AWS_EXTERNAL_ID   → <snowflake-external-id>
```

Step 2 — Create the IAM role with the values from DESC INTEGRATION:

Trust policy:

```
{
  "Version": "2012-10-17",
  "Statement": [{
    "Effect": "Allow",
    "Principal": {
      "AWS": "<snowflake-iam-user-arn>"
    },
    "Action": "sts:AssumeRole",
    "Condition": {
      "StringEquals": {
        "sts:ExternalId": "<snowflake-external-id>"
      }
    }
  }]
}
```

Permission policy — same S3 read/write as above, scoped to your bucket.

> If you ever recreate the Storage Integration, the `STORAGE_AWS_EXTERNAL_ID` changes. Update the IAM trust policy with the new value or Snowflake will lose S3 access.

Step 3 — Create the external stage:

```
CREATE OR REPLACE STAGE <your-snowflake-database>.<your-snowflake-schema>.
<your-stage>
  URL = 's3://<your-s3-bucket>/'
  STORAGE_INTEGRATION = <your-storage-integration>
  FILE_FORMAT = (TYPE = PARQUET);
```

EventBridge Scheduler (`<your-eventbridge-role>`)

Trust policy — `aws:SourceAccount` is mandatory:

```
{
  "Version": "2012-10-17",
  "Statement": [{
    "Effect": "Allow",
    "Principal": {"Service": "scheduler.amazonaws.com"},
    "Action": "sts:AssumeRole",
    "Condition": {
      "StringEquals": {"aws:SourceAccount": "<your-aws-account-id>"}
    }
  }]
}
```

> Without `aws:SourceAccount`, the schedule appears healthy in the console but fires zero invocations. This is the most common scheduler setup mistake.

Permission policy:

```
{
  "Version": "2012-10-17",
  "Statement": [{
    "Effect": "Allow",
    "Action": [
      "redshift-data:ExecuteStatement",
      "redshift:GetClusterCredentials"
    ],
    "Resource": "*"
  }]
}
```

Type Mapping: Snowflake → Redshift via Parquet

This is the most common source of `COPY` failures when moving data from Snowflake to Redshift. Reference this table when designing tasks for any table.

Snowflake Type	Parquet Type Written	Redshift Compatible Type	Notes
`NUMBER` / `NUMBER(p,0)`	`decimal128(38,0)`	`DECIMAL(38,0)`	Or cast `::BIGINT` → writes `int64`
`NUMBER` cast `::BIGINT`	`int64`	`BIGINT`	Preferred for integer IDs and counts
`NUMBER(p,s)` where s>0	`decimal128(p,s)`	`DECIMAL(p,s)`	Match precision and scale exactly
`VARCHAR`	`byte_array` (UTF-8)	`VARCHAR(n)`	Compatible directly
`TIMESTAMP_NTZ`	`timestamp[ms, tz=UTC]`	`TIMESTAMP`	Compatible directly
`DATE`	`date32`	`DATE`	Compatible directly
`BOOLEAN`	`bool`	`BOOLEAN`	Compatible directly

Key rule: Any Snowflake `NUMBER` that should land in a Redshift integer column (`INT`, `BIGINT`, `SMALLINT`) must be explicitly cast in the task SELECT:

```
-- In task SELECT — forces int64 Parquet output instead of decimal128
some_id::BIGINT    AS some_id,
some_count::BIGINT AS some_count
```

Verification Checklist

Flow 1: Redshift → Snowflake (shipments)

```
-- 1. Confirm watermark is advancing after each run (Redshift)
SELECT * FROM etl_watermarks WHERE table_name = 'shipments';

-- 2. Confirm UNLOAD wrote a new timestamped file (AWS CLI)
-- aws s3 ls s3://<your-s3-bucket>/redshift-to-snowflake/shipments/ --
recursive

-- 3. Confirm Snowpipe ingested the file (Snowflake)
SELECT * FROM TABLE(INFORMATION_SCHEMA.COPY_HISTORY(
    TABLE_NAME => 'shipments_staging',
    START_TIME => DATEADD('hour', -1, CURRENT_TIMESTAMP)
));

-- 4. Confirm staging has the new rows
SELECT shipment_id, order_id, carrier, status
FROM <your-snowflake-database>.<your-snowflake-schema>.shipments_staging
ORDER BY shipment_id DESC LIMIT 5;

-- 5. Run MERGE, then verify target count
SELECT COUNT(*) FROM <your-snowflake-database>.<your-snowflake-
schema>.shipments;
```

Flow 2: Snowflake → Redshift (inventory_forecast)

```
-- 1. Confirm stream has pending data after an insert/update (Snowflake)
SELECT SYSTEM$STREAM_HAS_DATA(
    '<your-snowflake-database>.<your-snowflake-
schema>.inventory_forecast_stream'
);

-- 2. Check task run history (Snowflake)
SELECT name, state, error_message, scheduled_time, completed_time
FROM TABLE(INFORMATION_SCHEMA.TASK_HISTORY(
    TASK_NAME => 'export_inventory_forecast_task',
    SCHEDULED_TIME_RANGE_START => DATEADD('hour', -24, CURRENT_TIMESTAMP)
))
ORDER BY scheduled_time DESC LIMIT 5;

-- 3. Confirm task wrote a Parquet file to S3 (AWS CLI)
-- aws s3 ls s3://<your-s3-bucket>/snowflake-to-
redshift/inventory_forecast/

-- 4. Verify rows landed in Redshift
-- SELECT COUNT(*) FROM inventory_forecast;
```

```
-- SELECT forecast_id, sku_id, metadata_action FROM inventory_forecast
ORDER BY forecast_id DESC LIMIT 5;
```

Phase-by-Phase Setup Order

Follow this order for a clean first deployment.

AWS Setup (one-time)

1. Create S3 bucket `<your-s3-bucket>`
2. Create IAM role `<your-redshift-s3-role>` (trust: `redshift.amazonaws.com`, permissions: S3 read/write on the bucket)
3. Attach `<your-redshift-s3-role>` to the Redshift cluster

Snowflake Setup (one-time)

4. Create Storage Integration `<your-storage-integration>`
5. Run `DESC INTEGRATION <your-storage-integration>` → copy `STORAGE_AWS_IAM_USER_ARN` and `STORAGE_AWS_EXTERNAL_ID`
6. Create IAM role for Snowflake S3 access using the ARN and ExternalId from step 5
7. Create external stage `<your-stage>` pointing at the S3 bucket

S3 Event Notification (one-time)

8. Create Snowpipe `shipments_pipe` → run `SHOW PIPES` → copy the `notification_channel` SQS ARN
9. In the S3 console, add an event notification on the bucket: prefix = `redshift-to-snowflake/shipments/`, destination = SQS ARN from step 8

Redshift Schema Setup (one-time)

10. Create `etl_watermarks` and insert the `shipments` row
11. Create `shipments` source table
12. Create `inventory_forecast` target table — **include all three metadata columns**
13. Create `inventory_forecast_staging` table (same schema as target)
14. Create `copy_inventory_forecast()` stored procedure
15. Create `export_shipments_incremental()` stored procedure

AWS Automation Setup (one-time)

16. Create IAM role `<your-eventbridge-role>` with trust: `scheduler.amazonaws.com` (include `aws:SourceAccount` condition) and permissions: `redshift-data:ExecuteStatement`, `redshift:GetClusterCredentials`
17. Create EventBridge schedule for RS→SF export: `CALL export_shipments_incremental()` on your cadence
18. Create EventBridge schedule for SF→RS COPY: `CALL copy_inventory_forecast()` at `:30` UTC hourly

Snowflake Schema Setup (one-time)

19. Create `shipments` target table and `shipments_staging` table
20. Create `shipments_staging_stream` on the staging table
21. Create and resume `merge_shipments_task`
22. Create `inventory_forecast` source table
23. Create `inventory_forecast_stream` on the source table
24. Create and resume `export_inventory_forecast_task`

Phase 1: Bulk Load (run once)

25. Redshift: UNLOAD full `shipments` table to the `_bulk/` S3 prefix
26. Snowflake: COPY bulk file into `shipments` target table
27. Snowflake: Insert seed data into `inventory_forecast` → `EXECUTE TASK` manually
28. Redshift: COPY from `snowflake-to-redshift/inventory_forecast/`
29. Reset the Redshift watermark: `UPDATE etl_watermarks SET last_exported_at = SYSDATE WHERE table_name = 'shipments'`

Phase 2: Incremental (ongoing — fully automated)

30. EventBridge schedule fires `CALL export_shipments_incremental()` on your configured cadence
31. Snowpipe auto-ingests new Parquet files into `shipments_staging`
32. `shipments_staging_stream` detects new rows → `merge_shipments_task` fires automatically (at most hourly) and MERGEs the delta into the target
33. `inventory_forecast_stream` feeds `export_inventory_forecast_task` hourly; Snowflake task writes Parquet to S3 at `:00` UTC
34. EventBridge schedule fires `CALL copy_inventory_forecast()` at `:30` UTC; procedure TRUNCATEs staging, COPYs Parquet, applies CDC (DELETE+INSERT) to target